19 At the start of day n, the number of cells being used in an experiment is T_n

At the start of the next day, the number of cells being used in the experiment is T_{n+1} where

$$T_{n+1} = kT_n \text{ and } k \text{ is a positive constant.}$$

Given that

$T_1 = 250\,000$

$T_3 = 490\,000$

$T_7 : T_4 = m : 1$

find the value of m.
Give your answer correct to 3 significant figures.

$T2 = 250000k$; $T3 = 250000k^2$.

1 mark
Use two recurrence steps.

$490000 = 250000k^2$

$k^2 = 490000 \div 250000 = 1.96$

1 mark
Find k^2.

$k > 0$, so $k = \sqrt{1.96} = 1.4$

$T7 = k^3 T4$, so $T7 \div T4 = k^3$

1 mark
Three steps from T4 to T7.

$m = 1.4^3 = 2.744$

1 mark
Evaluate k^3.

$m = 2.74$ (3 significant figures)

1 mark
2.74.

(Total for Question 19 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA